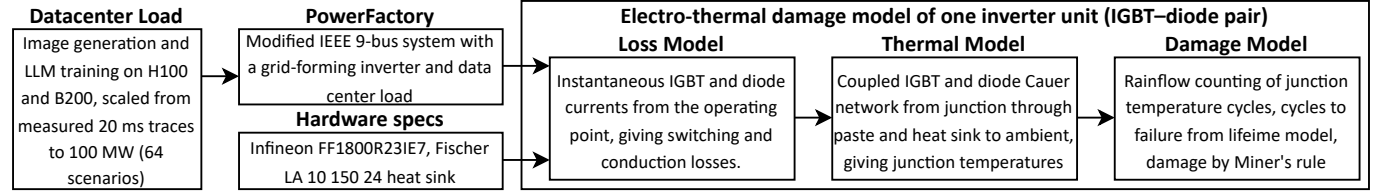
Datacenter Load
Image generation and LLM training on H100 and B200, scaled from measured 20 ms traces to 100 MW (64 scenarios)
PowerFactory
Modified IEEE 9-bus system with a grid-forming inverter and data center load
Hardware specs
Infineon FF1800R23IE7, Fischer LA 10 150 24 heat sink
Electro-thermal damage model of one inverter unit (IGBT–diode pair)
Loss Model
Instantaneous IGBT and diode currents from the operating point, giving switching and conduction losses.
Thermal Model
Coupled IGBT and diode Cauer network from junction through paste and heat sink to ambient, giving junction temperatures
Damage Model
Rainflow counting of junction temperature cycles, cycles to failure from lifeime model, damage by Miner's rule